PROJECT SYSTEM ARCHITECTURE OVERVIEW

Data Flow Example

video.mp4

Frame Processing

Frame 1
→ No motion
→ Skip
→ dummy result

Frame 2
→ Motion detected
→ Check rules
• FireBlockYCbCrRule → PASS
• Crop ROI
• Run CNN
• pred='fire_smoke'

Result Output

CSV: video_results.csv
(frame_idx, pred, probs, time)

Video: video_out. mp4
• Draw pred label + FPS
• Draw block grid
• Highlight fire blocks

Metric Calculation

Results:
• Accuracy: 0.95
• F1-Score: 0.92
• FPS: 30.5
• FPR: 0.03

Design Patterns Used

- Factory Pattern: MethodFactory, SkipProcFactory, MetricSrcFactory
- Strategy Pattern: BaseSkipProc (different skip strategies)
- Observer/Handler Pattern: BaseRsProc (multiple result handlers)
- Template Method: BaseMethod (defines inference workflow)
- Adapter Pattern: BaseCsvLoader (dataset-specific parsing)
- Composite Pattern: BaseRule (AnyRule, AllRule)
- Pipeline Pattern: VideoPipeline + Renderers

Config. py

- DatasetCfg (video paths, labels)
- MethodCfg (which method to use)
- MetricSetCfg (which metrics to compute)
- ModelConfig (CNN model info)
- InferConfig (save_csv, save_video, etc.)

configures

Paper2Exp
Experiment Orchestrator

1. Creates Method

src/methods/

MethodFactory

creates

BaseMethod (ABC)

- NoTempMethod (frame-by-frame)
- TempMethod (with skip)
- TempBaselineTptMethod (temporal filtering)

uses

Skip Pipeline

SkipProcFactory

creates

BaseSkipProc

- BlockSkipProc (rule-based detection)
- ProfSkipProc (motion ROI cropping)
- NoSkipProc (no optimization)
- RandSkipProc (random skip)

uses

uses

Motion Components

- FrameDiffDet
- AccMotionDet
- FireBlockYCbCrRule
- WaveletRule
- etc.

2. Inference on Videos

src/results/

BaseRsProc (ABC)

CsvRSProc

Saves:
video_name_results.csv
Columns:
• frame_idx
• pred_label
• probs
• logits
• elapsed_time

VideoInferRsProc

VideoPipeline

Multiple Renderers

- InferRsRenderer (OSD: pred, fps)
- GridRenderer (block grid)
- BlockRuleRenderer (fire/smoke)
- BlockProfRenderer (motion blocks)

Subclasses

- VideoBlockSkipRsProc
- FgmaskBlockSkipRsProc (visualize motion masks)

saves

saves

Output Files

- video1_results.csv
- video2_results.csv
- video1_out.mp4
- video2_out.mp4

3. Metric Calculation

src/metrics/

MetricSrcFactory

creates

BaseMetricSrc (ABC)

CsvMetricSrc

CSV Loaders

- BaseCsvLoader (ABC)
- DFireCsvLoader
- UFireIndoorCsvLoader

Process

1. Load *_results.csv
2. Load ground truth
3. Match pred vs GT

passes data

TorchMetrics

- FPS (frames per second)
- FPR (false positive rate)
- Accuracy
- F1Score
- Precision
- Recall (TPR)

produces

performance_metrics.csv

Key Interactions

1. CONFIG → METHODS:
Config selects method & provides model info

2. METHODS → RESULTS:
Handlers save CSV and/or video

3. RESULTS → METRICS:
Load CSV + ground truth → compute metrics

4. SKIP PIPELINE:
Motion detection → Rule checking → Inference